

Student ID: 285269

Homework No 1

Problem 1 Solid Angle:

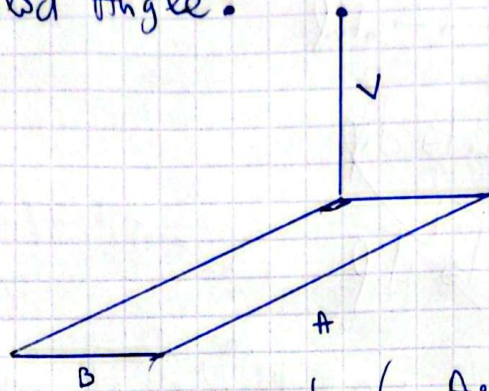
$$V = 12,2 \text{ m}$$

$$A = 3,9 \text{ m}$$

$$B = 1,1 \text{ m}$$

$$d\Omega = \frac{dS \cdot \epsilon}{r^2}$$

I couldn't get derivative myself



$$\Omega = \arctan \left(\frac{A \cdot B}{V \cdot \sqrt{V^2 + A^2 + B^2}} \right) \approx 0,0277 \text{ steradians.}$$

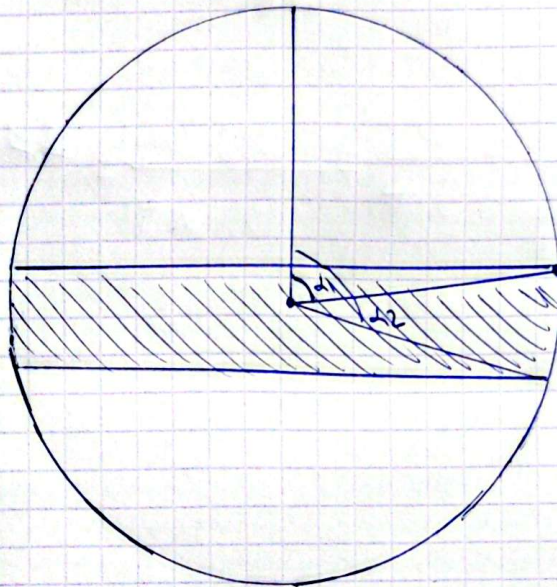
Problem 2

$$I = 300 \text{ cd}$$

$$\alpha_1 = 86^\circ$$

$$\alpha_2 = 116^\circ$$

$$d\Omega = \sin \theta d\theta d\varphi$$



$$[d\Omega_1 = \sin \alpha_1 d\alpha_1 d\varphi]$$

$$\Omega_1 = \int_0^{2\pi} \int_0^{86} \sin \alpha_1 d\alpha_1 d\varphi \rightarrow \left[-\cos \alpha_1 \right]_0^{86} = -\cos 86 + 1$$

$$\Omega_1 = \int_0^{2\pi} (-\cos 86 + 1) d\varphi = 2\pi (-\cos 86 + 1)$$

$$\Omega_2 = 2\pi (-\cos 116 + 1)$$

$$\Omega_2 - \Omega_1 = 2\pi (-\cos 116 + 1 + \cos 86 - 1) = 2\pi (\cos 86 - \cos 116) =$$

$$= 3,19 \text{ sr}$$

$$I = \frac{\Phi}{\Omega}$$

$$\left[\Phi = I \cdot \Omega = 957,3 \text{ lm} \right]$$

Homework 2

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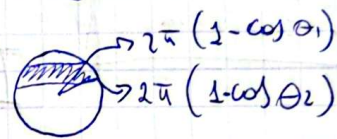
Problem 3:

$$\Phi = 120 \text{ kV} \quad I_{\text{mean}} = \frac{\Phi}{\Omega_{\text{total}}}$$

$$I_{\text{mean}} = ? \quad * d\Omega = \sin \theta \, d\theta \, d\varphi$$

$$\Omega = \int_0^{2\pi} \int_0^{\theta} \sin \theta \, d\theta \, d\varphi = \int_0^{2\pi} [-\cos \theta]_0^{\theta} d\varphi =$$

$$= \int_0^{2\pi} (-\cos \theta + 1) d\varphi = 2\pi (1 - \cos \theta)$$



Steps: $A_1 - A_2$; $A_3 - A_4$; $A_5 - A_6$

$$\Omega_{A_1, A_2} = 2\pi (\cos 13 - \cos 18) = 0,146 \text{ sr}$$

$$\Omega_{A_3, A_4} = 2\pi (\cos 51 - \cos 56) = 0,44 \text{ sr}$$

$$\Omega_{A_5, A_6} = 2\pi (\cos 77 - \cos 82) = \cancel{0,146} 0,53 \text{ sr}$$

$$\left[I_{\text{mean}} = \frac{\Phi}{\Omega_{\text{total}}} = \frac{120}{\frac{0,146 + 0,44 + 0,53}{1,116}} = \cancel{107,52} 107,52 \text{ dV} \right]$$

Problem 4

$I_{\text{menyapleskal}} = ?$

$$P = 150 \text{ W}$$

$$\Phi = 7,5 \text{ klm}$$

$$\eta = ?$$

$$\tau = 0,64$$

$$D = 0,65 \text{ m}$$

$$\Phi = P \cdot \eta$$

$$\left[\eta = \frac{\Phi}{P} = \frac{7,5 \times 10^3}{150} = 50 \text{ lm/W} \right]$$

$$\Phi_{\text{ex}} = \Phi \cdot \tau = 4800 \text{ lm}$$

$$\left[I_{\text{menyapleskal}} = \frac{\Phi}{4\pi} = \frac{7500}{4\pi} = 596,8 \text{ cd} \right]$$

$$A = 4\pi R^2 = 1,327 \text{ m}^2$$

$$H = \frac{\Phi_{\text{ex}}}{S} \rightarrow \left[L = \frac{\Phi_{\text{ex}}}{S \cdot \pi} = \frac{4800 \text{ lm}}{1,327 \times \pi} = 1,151 \cdot 10^3 \frac{\text{cd}}{\text{m}^2} \right]$$